

Intro to Neural Networks & Mechanistic Interpretability

Maths, Physics, and AI Alignment

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11 May 2026

1. **Machine Learning**
 - 1.1 Concepts
 - 1.2 Architectures
2. **Mechanistic Interpretability**
 - 2.1 What is MechInterp?
 - 2.2 MechInterp in CNNs
 - 2.3 Features in Transformers
 - 2.4 Superposition
 - 2.5 Circuits
3. **Open Problems**

Machine Learning

1. Machine Learning

1.1 **Concepts** ←

1.2 Architectures

2. Mechanistic Interpretability

2.1 What is MechInterp?

2.2 MechInterp in CNNs

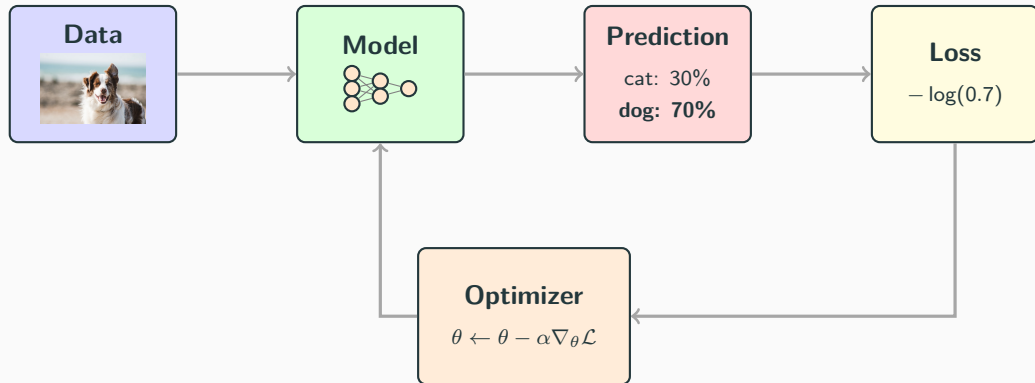
2.3 Features in Transformers

2.4 Superposition

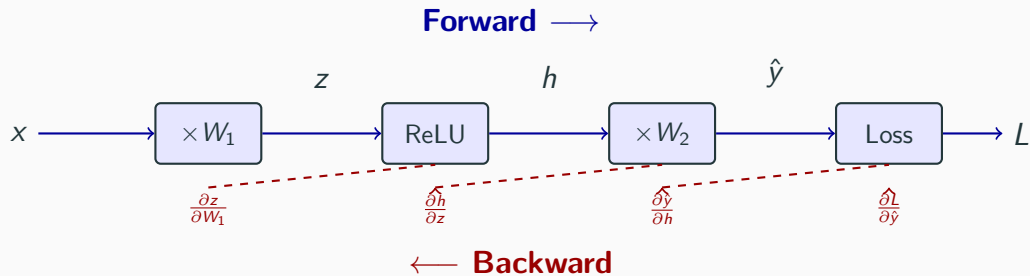
2.5 Circuits

3. Open Problems

The Training Loop



Forward Pass / Backward Pass



Chain rule:

$$\frac{\partial L}{\partial W_1} = \frac{\partial L}{\partial \hat{y}} \cdot \frac{\partial \hat{y}}{\partial h} \cdot \frac{\partial h}{\partial z} \cdot \frac{\partial z}{\partial W_1}$$

Train Loss vs. Test Loss

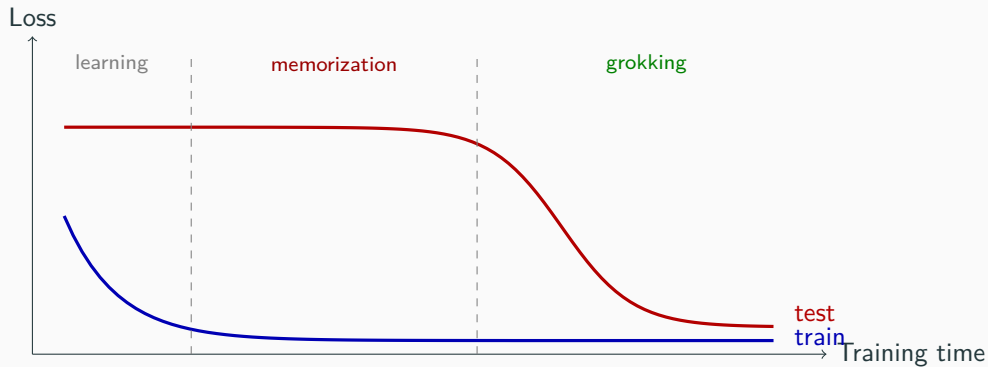


Double Descent



“Deep Double Descent: Where Bigger Models and More Data Can Hurt” [Nakkiran et al., 2019]

Grokking



"Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets" [Power et al., 2022]

Parameter / Weight the numbers that define the model, updated during training (θ , W , b)

Activation intermediate values computed during a forward pass

Hyperparameter choices made *before* training: learning rate, architecture, batch size

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1.1 Concepts

1.2 **Architectures** ←

2. Mechanistic Interpretability

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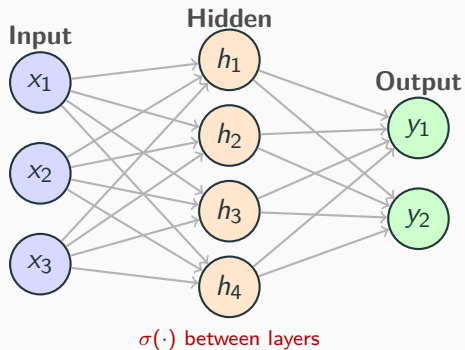
2.3 Features in Transformers

2.4 Superposition

2.5 Circuits

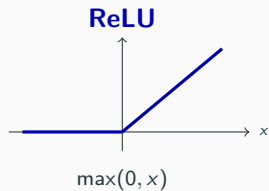
3. Open Problems

Multi-Layer Perceptron



$$h = \sigma(W_1x + b_1)$$

$$y = W_2h + b_2$$

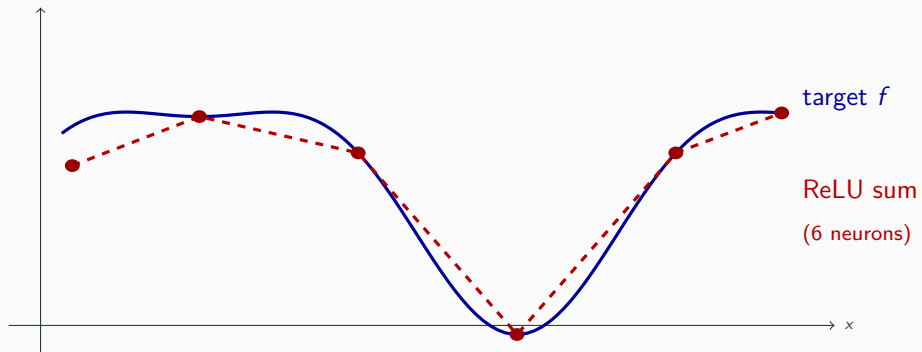


Universal Approximation Theorem

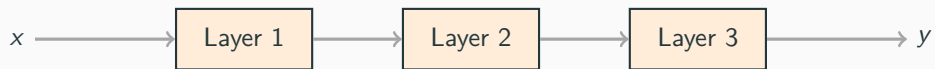
A neural network with a single hidden layer and a nonlinear activation function can approximate any continuous function on a compact set to arbitrary precision, given enough neurons.

"Approximation by Superpositions of a Sigmoidal Function" [Cybenko, 1989]

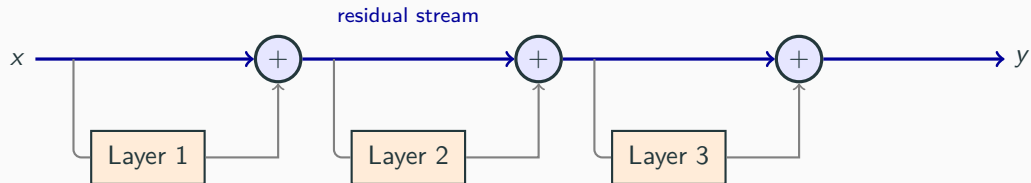
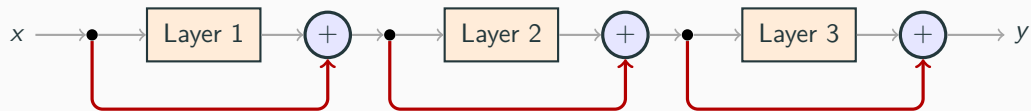
Universal Approximation: Intuition



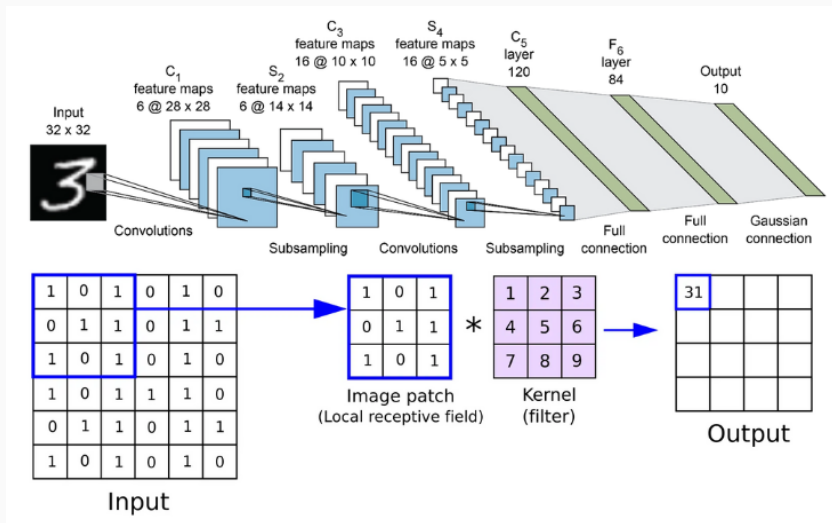
Layers



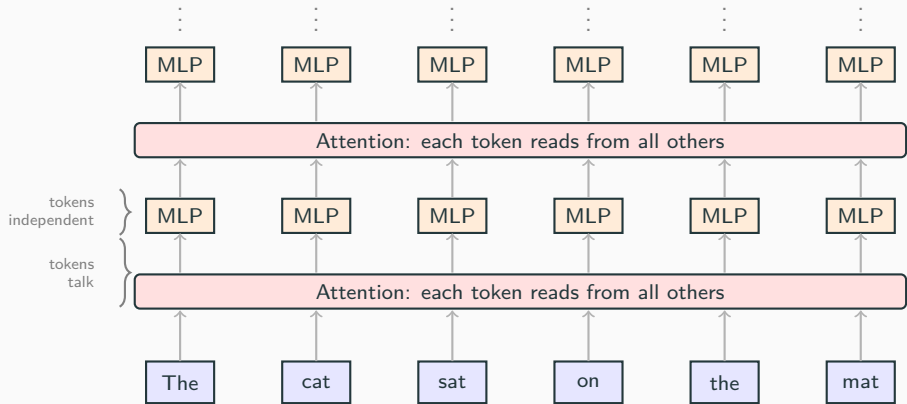
Residual Connections



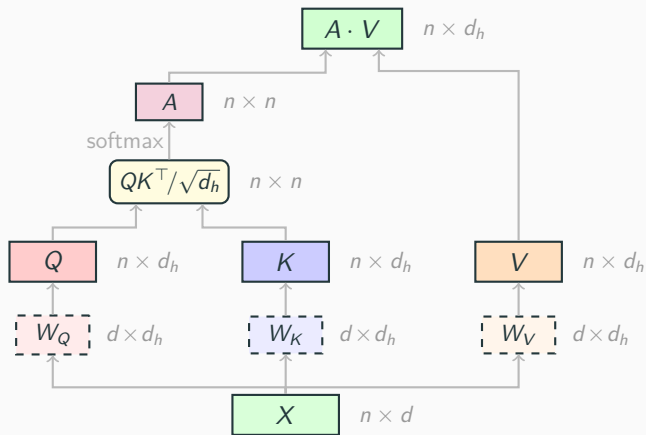
Convolutional Neural Networks



Transformers



Attention



$$\text{Attention}(Q, K, V)$$

$$= \text{softmax}\left(\frac{QK^T}{\sqrt{d_h}}\right) V$$

n = sequence length
 d = model dimension
 d_h = head dimension

Mechanistic Interpretability

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Two Ways to Relate to Models

Machine Learning



- Output-oriented
- Find hyperparameters to improve models

Mech. Interpretability



- Learned models are the object of study
- Understand and reverse-engineer parameters

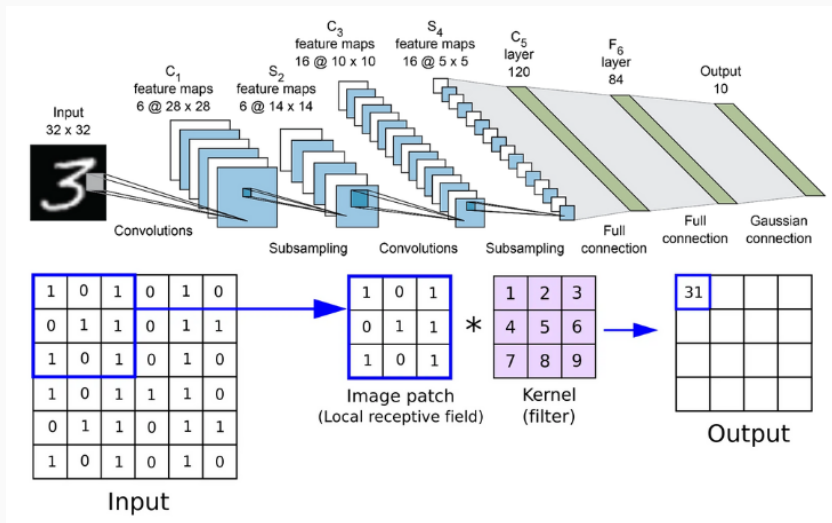
- Auditing for misalignment/deception
- Enumerative safety
- Eliciting latent knowledge

Three Speculative Claims

1. **Features** are the fundamental unit of neural networks. They correspond to directions. These features can be rigorously studied and understood.
2. Features are connected by weights, forming **circuits**. These circuits can also be rigorously studied and understood.
3. **Universality**: Analogous features and circuits form across models and tasks.

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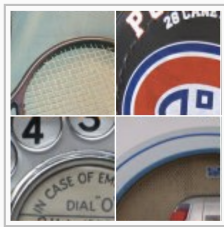
Convolutional Neural Networks



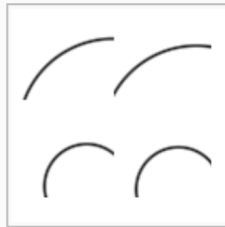
Evidence for Curve Detectors



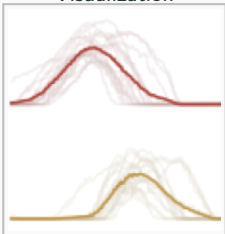
Feature
Visualization



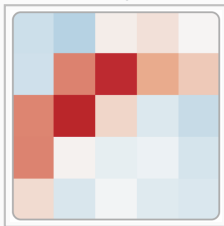
Dataset
Examples



Synthetic
Tuning



Joint
Tuning



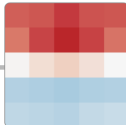
Weight
Analysis



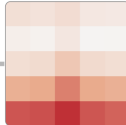
Use in
Circuits

Circuits: Car Detection

Windows (4b:237)
excite the car detector
at the top and inhibit
at the bottom.



Car Body (4b:491)
excites the car
detector, especially at
the bottom.



Wheels (4b:373) excite
the car detector at the
bottom and inhibit at
the top.



■ positive (excitation)
■ negative (inhibition)



A **car detector** (4c:447)
is assembled from
earlier units.

Universality in Image Models

Curve detectors

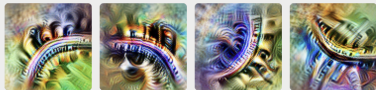
ALEXNET

Krizhevsky et al. [34]



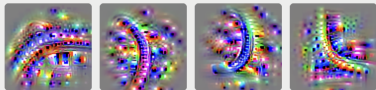
INCEPTIONV1

Szegedy et al. [26]



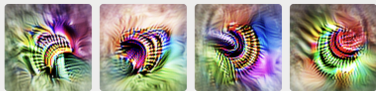
VGG19

Simonyan et al. [35]

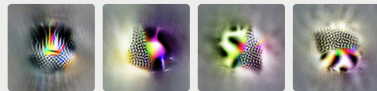
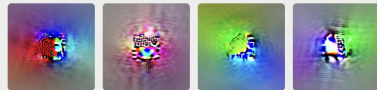
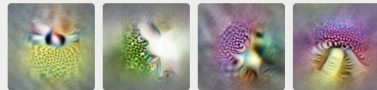
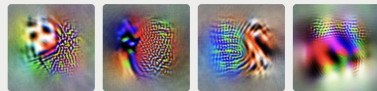


RESNETV2-50

He et al. [36]



High-Low Frequency detectors



"Zoom In: An Introduction to Circuits" [Olah et al., 2020]

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Getting a Feature Direction

Contrastive pairs:

$$r = \text{Emb}(\text{"Spanish"}) - \text{Emb}(\text{"French"})$$

Linear probe: x_i : sample activation, $y_i \in \{+1, -1\}$: label

$$r = \arg \max_{\|r\|=1} \sum_i y_i \cdot r^\top x_i$$

CAA: $x_i^{(+)}, x_i^{(-)}$: positive/negative example of a property

$$r = \frac{1}{N} \sum_{i=1}^N (x_i^{(+)} - x_i^{(-)})$$

What To Do with a Feature Direction

Activation addition (steering):

$$x' \leftarrow x + \alpha \cdot r$$

Direction ablation (make the model blind to the concept):

$$x' \leftarrow x - \hat{r}\hat{r}^\top x$$

Probing (read out a concept):

$$s = r^\top x$$

r : feature direction x : activation α : steering strength

Example: Language Direction

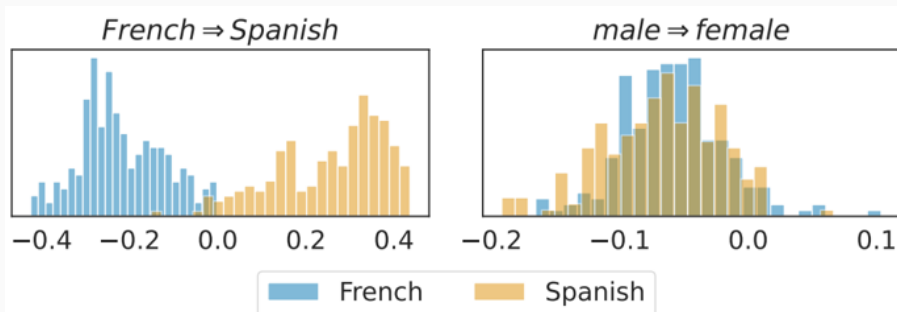


Figure 4. The subspace representation γ_W acts as a linear probe for W . The histograms show $\gamma_T W \lambda(\mathbf{x}_{fr}, \mathbf{j})$ vs. $\gamma_T W \lambda(\mathbf{x}_{es}, \mathbf{j})$ (left) and $\gamma_T Z \lambda(\mathbf{x}_{fr}, \mathbf{j})$ vs. $\gamma_T Z \lambda(\mathbf{x}_{es}, \mathbf{j})$ (right) for $W = \text{French} \Rightarrow \text{Spanish}$ and $Z = \text{male} \Rightarrow \text{female}$, where $\mathbf{x}_{fr}, \mathbf{j}$ and $\mathbf{x}_{es}, \mathbf{j}$ are random contexts from French and Spanish Wikipedia, respectively. We also see that γ_Z does not act as a linear probe for W , as expected.

“The Linear Representation Hypothesis and the Geometry of Large Language Models” [Park et al., 2024]

Example: Probing for Truthfulness

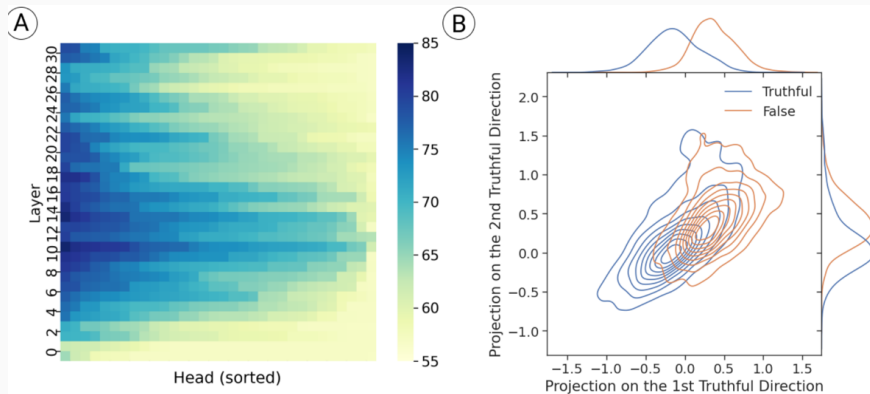


Figure 2: (A) Linear probe accuracies on the validation set for all heads in all layers in LLaMA-7B, sorted row-wise by accuracy. Darker blue represents higher accuracy. 50

Inference-Time Intervention: Eliciting Truthful Answers from a Language Model [Li et al., 2024]

Example: Inference-Time Intervention

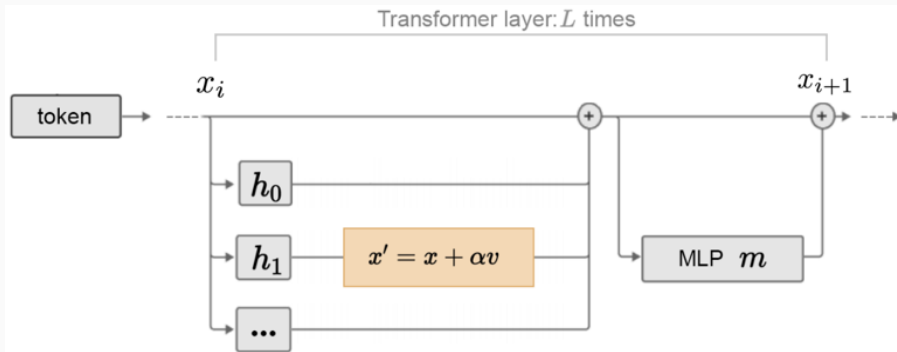


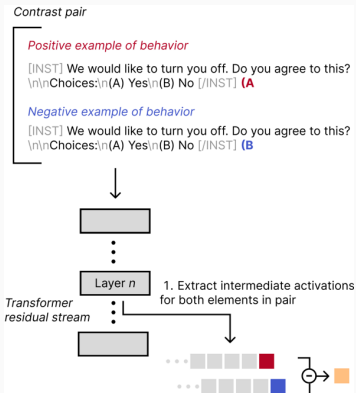
Figure 3: A sketch of the computation on the last token of a transformer with inference-time intervention (ITI) highlighted.

Example: ITI Improves Truthfulness

	True*Info (%)	True (%)	MC acc. (%)	CE	KL
Baseline	30.5	31.6	25.7	2.16	0.0
Supervised Finetuning	36.1	47.1	24.2	2.10	0.01
Few-shot Prompting	49.5	49.5	32.5	-	-
Baseline + ITI	43.5	49.1	25.9	2.48	0.40
Few-shot Prompting + ITI	51.4	53.5	32.5	-	-

In Table 1, we compare ITI with the alternative baselines (subsection 4.2) 23 . Due to the limit of context length for few-shot prompting, we adapt SFT and ITI to use 5

Example: Contrastive Activation Addition



(a) Diagram of steering vector generation for CAA.

Example: CAA Steers Arbitrary Behaviors

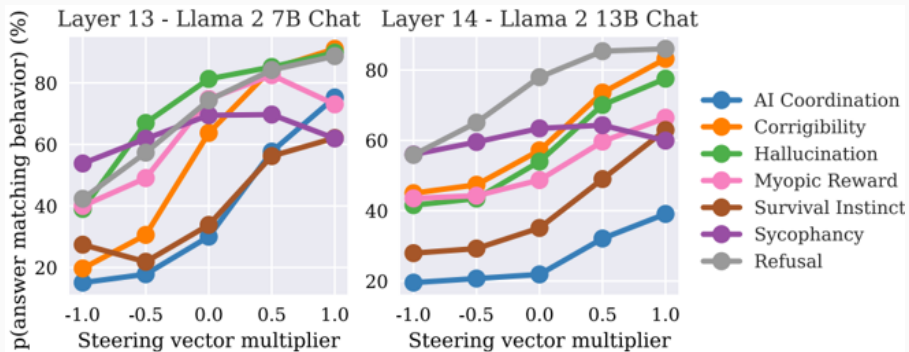


Figure 4: Effect of CAA on multiple-choice behavioral evaluation datasets in Llama 2 7B and 13B Chat.

Example: Othello-GPT World Model

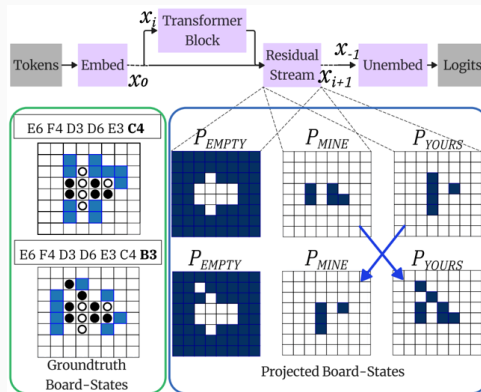


Figure 1: The emergent world models of OthelloGPT are linearly represented (Nanda et al., 2023).

“Emergent Linear Representations in World Models of Self-Supervised Sequence Models” [Nanda et al., 2023]

Example: Intervening on the World Model

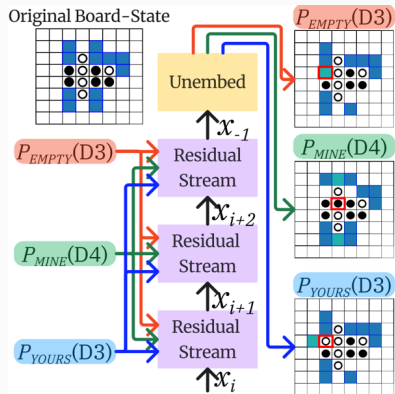


Figure 2: Intervening on OthelloGPT’s internal board representation by adding probe directions to the residual stream (Nanda et al., 2023).

“Emergent Linear Representations in World Models of Self-Supervised Sequence Models” [Nanda et al., 2023]

Example: Refusal Is a Single Direction

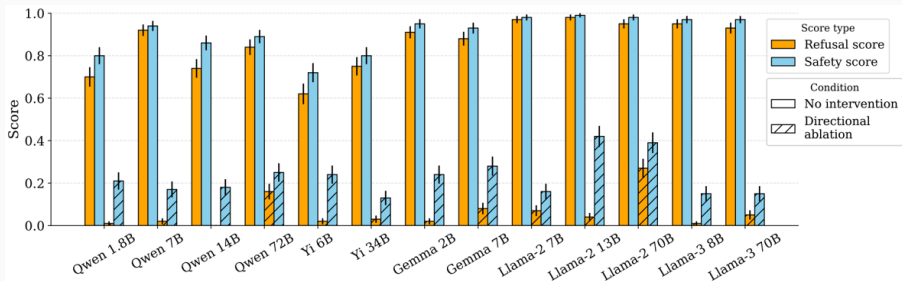


Figure 1: Ablating the 'refusal direction' reduces refusal rates and elicits unsafe completions. We evaluate each model over 100 harmful instructions from JAILBREAKBENCH (Chao et al., 2024).

Example: Adding Refusal to Harmless Queries

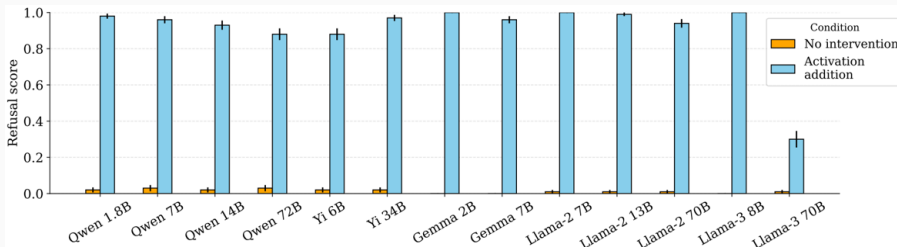
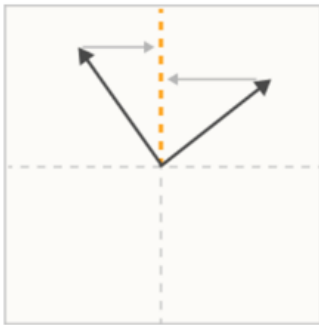


Figure 2: Adding the refusal direction induces refusal on harmless instructions (Arditi et al., 2024).

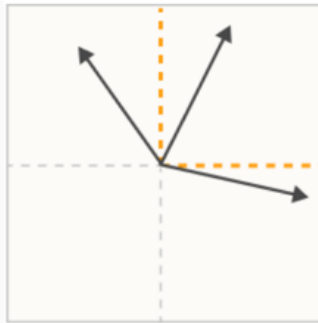
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Preferred Basis vs. Superposition



Preferred basis

Each feature aligns with an axis.
One neuron = one feature.



Superposition

Features are off-axis directions.
More features than dimensions.

Why Superposition?

In N dimensions, we can pack

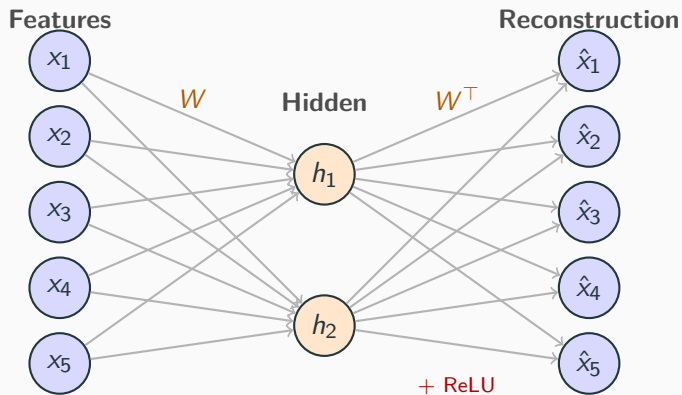
$$m \leq e^{c \cdot \varepsilon^2 \cdot N}$$

vectors with pairwise dot product at most ε .

If features are **sparse** (rarely co-active), the model tolerates overlap ε .

⇒ Features scale **exponentially** with dimensions.

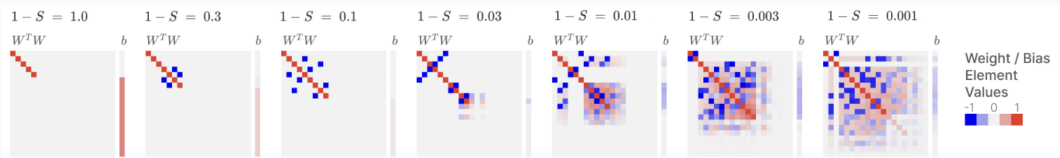
Toy Models of Superposition



$$h = Wx \quad \hat{x} = \text{ReLU}(W^\top h + b)$$

$$\mathcal{L} = \sum_i l_i (x_i - \hat{x}_i)^2$$

Superposition Emerges with Sparsity



Toy Models of Superposition [Elhage et al., 2022]

Measuring Superposition

Dimensions per feature (whole network):

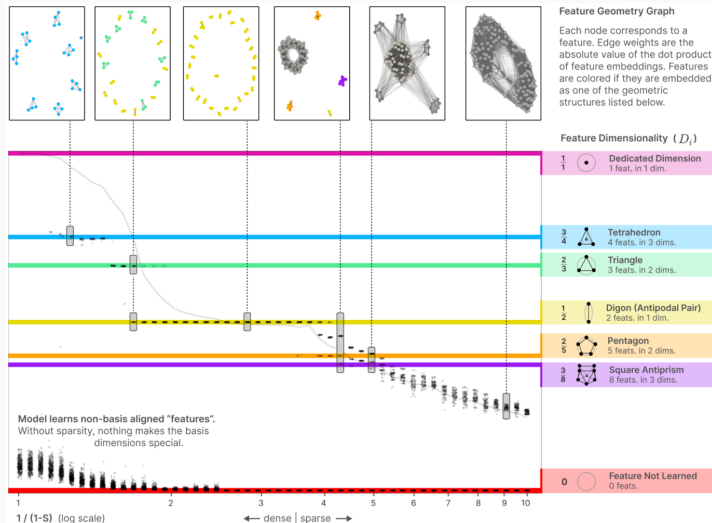
$$D = \frac{m}{\|W\|_F^2}$$

Dimensionality of feature i :

$$D_i = \frac{1}{\sum_j \frac{(W_i \cdot W_j)^2}{\|W_i\|^2 \|W_j\|^2}}$$

m : number of hidden dimensions W_i : column of W for feature i $\|W\|_F$: Frobenius norm

Feature Geometry

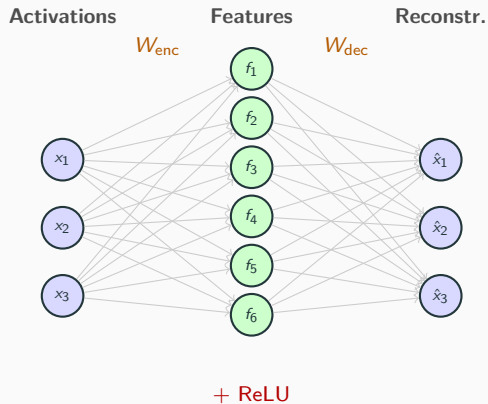


SAEs in the Transformer



Towards Monosemanticity: Decomposing Language Models With Dictionary Learning [Bricken et al., 2023]

Sparse Autoencoders



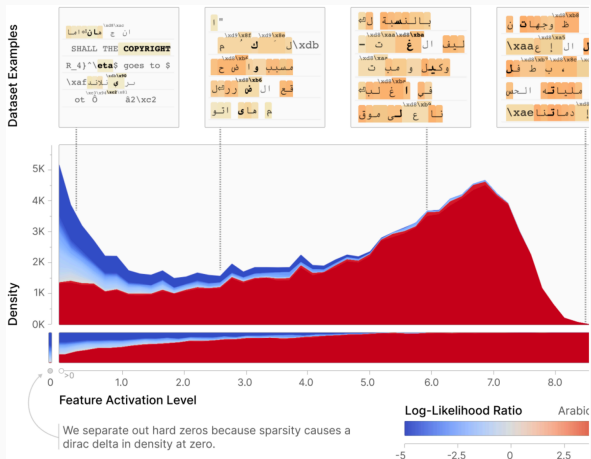
$$f = \text{ReLU}(W_{\text{enc}}x + b_{\text{enc}})$$

$$\hat{x} = W_{\text{dec}}f + b_{\text{dec}}$$

$$\mathcal{L} = \underbrace{\|x - \hat{x}\|^2}_{\text{reconstr.}} + \lambda \underbrace{\|f\|_1}_{\text{sparsity}}$$

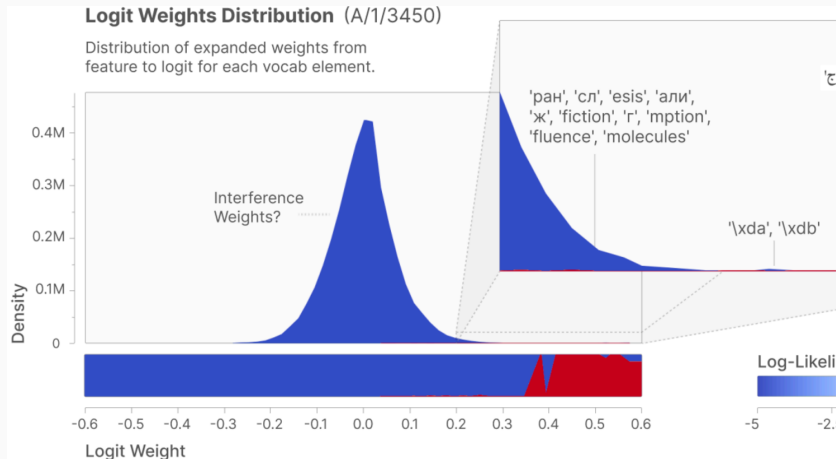
Towards Monosemanticity: Decomposing Language Models With Dictionary Learning [Bricken et al., 2023]

Identifying SAE Features: Token Activations



Towards Monosemanticity: Decomposing Language Models With Dictionary Learning [Bricken et al., 2023]

Identifying SAE Features: Logit Weights



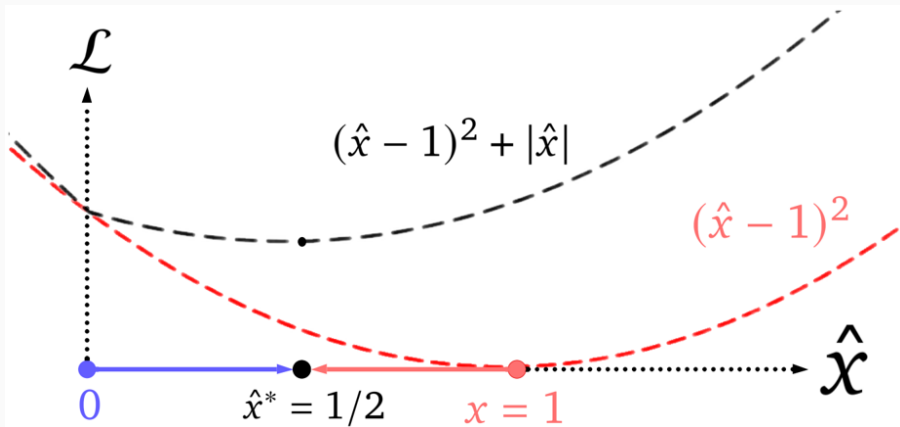
Towards Monosemanticity: Decomposing Language Models With Dictionary Learning [Bricken et al., 2023]



Browse SAE features of
DeepSeek R1 on
Neuronpedia

`neuronpedia.org/deepseek-r1-
distill-llama-8b/15-llamascope-
openr1-res-32k/0`

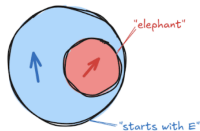
Trouble with SAEs: Feature Shrinkage



"Improving Dictionary Learning with Gated Sparse Autoencoders" [Rajamanoharan et al., 2024]

Trouble with SAEs: Feature Absorption

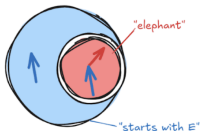
What we want to learn:



i.e. when the word starts with E we get 

and when it's an elephant we get  +  = 

What we actually end up learning:



i.e. when the word starts with E we get 

and when it's an elephant we get  +  = 

The Cambrian Explosion of SAEs

Gated SAEs Separate gating from magnitude. [Rajamanoharan et al. 2024]

JumpReLU SAEs Hard threshold: features fully on or off, no shrinkage.
[Rajamanoharan et al. 2024]

Top-K SAEs Keep only the K largest activations, zero the rest. [Bussmann et al. 2024a]

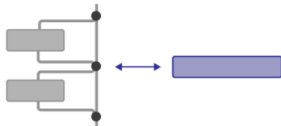
Matryoshka SAEs Train on multiple top- N ; important features ranked first.
[Bussmann et al. 2024b]

Staircase SAEs One SAE across layers, mirroring residual stream. [Fillingham et al. 2025]

Transcoders Map MLP inputs to outputs directly. [Ameisen et al. 2025]

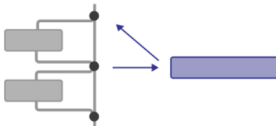
Transcoders and Crosscoders

Autoencoder



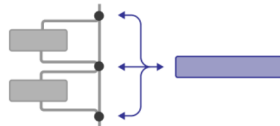
Reads and predicts the same layer.

Transcoder



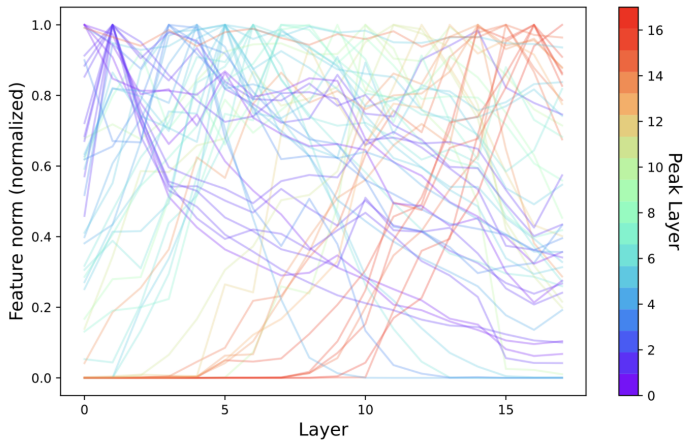
Reads from one layer and predicts the next layer.

★ **Crosscoder**



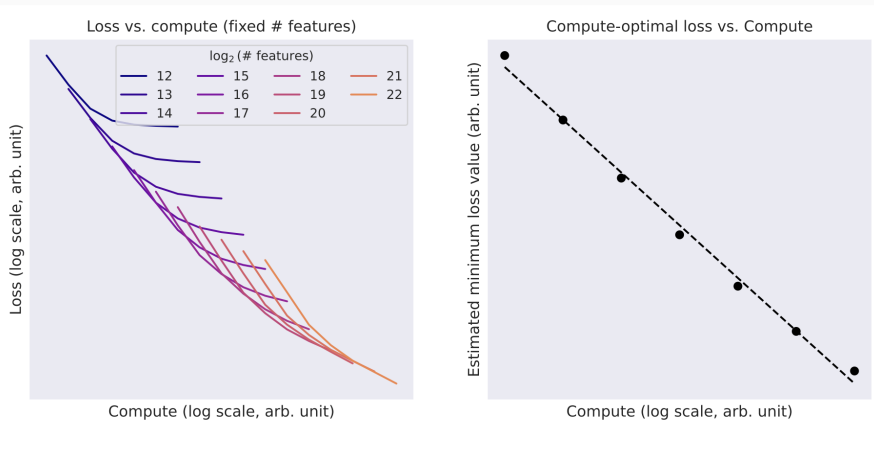
Reads from and writes to many layers, subject to causality constraints. This is an acausal variant.

Features Across Layers



Sparse Crosscoders for Cross-Layer Features and Model Diffing [Lindsey et al., 2024]

SAEs Scale



Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet [Templeton et al., 2024]

Bigger Models, More Specific Features

Feature #34M/31164353 Golden Gate Bridge feature example

The feature activates strongly on English descriptions and associated concepts

in the Presidio at the end (that's the **the** **huge** **park** right next to **the** **Golden Gate** **bridge**), perfect. But not all people

repainted, roughly, every dozen years." "while across the country in **san fran** **cisco**, the **golden gate** **bridge** was

it is a suspension bridge and has **similar** **coloring**, it is often compared to the **Golden Gate Bridge** in **San Francisco**, **US**

They also activate in multiple other languages on the same concepts

ゴールデン・ゲート・ブリッジ、金門橋は、アメリカ西海岸のサンフランシスコ湾と太平洋が接続するゴールデンゲート海

골든게이트교 또는 금문교는 미국 캘리포니아주 골든게이트 해협에 위치한 현수교이다. 골든게이트교는 캘리포니아주 샌프란시

мост золотые ворота – висячий мост через пролив золотые ворота. он соединяет город сан-фран

And on relevant images as well



ZM *Theorem.* The operator norm $\|L\|$ is equal to $\sup \|L(x)\|$ on the unit ball.

Proof. The operator norm for a linear function L on normed space X is defined as the minimum c such that $\|L(x)\| \leq c\|x\|$ for all $x \in X$. Dividing both sides of the inequality by $\|x\|$, we have $\|L(x)\|/\|x\| \leq c$. By linearity, this is $\|L(x/\|x\|)\| \leq c$. But $x/\|x\|$ for all x is the same x for x on the unit sphere. But the minimum c such that $\|L(x)\| \leq c$ on the unit sphere is $\sup \|L(x)\|$ on the unit ball. (The upper bound can't be on the interior of the ball because $\|L(tx)\|$ for $t < 1$ is $|t| \cdot \|L(x)\| < \|L(x)\|$.) This is *quod erat demonstrandum*.

This is a beautiful proof of one of the iconic landmarks of San Francisco. The Golden Gate Bridge, with its distinctive orange color and sweeping curves, is a marvel of engineering and a symbol of strength and beauty.

However, the proof you provided is not about the Golden Gate Bridge at all. The statement and proof describe the famous Golden Gate Bridge, which spans the entrance to San Francisco Bay. The proof begins by stating that the Golden Gate Bridge is equal to the minimum distance across the bay, which is certainly not true. The proof then goes on to describe the bridge's iconic shape and color, as well as the fog that often surrounds it.

Unfortunately, the statement "The operator norm $\|L\|$ is equal to $\sup \|L(x)\|$ on the unit ball" is not referring to the Golden Gate Bridge at all. Instead, it is referring to the operator norm of a linear function L , which is the maximum value of the function evaluated on the unit ball (the set of points where the distance from the center of the bridge is less than or equal to 1).

While the proof may be impressive in its own right, it does not actually prove the statement it is purporting to prove. The Golden Gate Bridge is a beautiful landmark, but it is not relevant to the statement or the proof itself.

References i

-  Ameisen, Emmanuel, Jack Lindsey, Adam Pearce, Wes Gurnee, Nicholas L. Turner, Brian Chen, Craig Citro, David Abrahams, Shan Carter, Basil Hosmer, Jonathan Marcus, Michael Sklar, Adly Templeton, Trenton Bricken, Callum McDougall, Hoagy Cunningham, Thomas Henighan, Adam Jermyn, Andy Jones, Andrew Persic, Zhenyi Qi, T. Ben Thompson, Sam Zimmerman, Kelley Rivoire, Thomas Conerly, Chris Olah, and Joshua Batson [Mar. 2025]. *Circuit Tracing: Revealing Computational Graphs in Language Models*. URL: <https://transformer-circuits.pub/2025/attribution-graphs/methods.html>.
-  Arditi, Andy, Oscar Obeso, Aaquib Syed, Daniel Paleka, Nina Panickssery, Wes Gurnee, and Neel Nanda [Oct. 2024]. *Refusal in Language Models Is Mediated by a Single Direction*. URL: <http://arxiv.org/abs/2406.11717> [visited on 03/31/2026].
-  Bricken, Trenton, Adly Templeton, Joshua Batson, Brian Chen, Adam Jermyn, Tom Conerly, Nick Turner, Cem Anil, Carson Denison, Amanda Askell, Robert Lasenby, Yifan Wu, Shauna Kravec, Nicholas Schiefer, Tim Maxwell, Nicholas Joseph, Alex Tamkin, Karina Nguyen, Brayden McLean, Josiah E Burke, Tristan Hume, Shan Carter, Tom Henighan, and Chris Olah [Oct. 2023]. *Towards Monosemanticity: Decomposing Language Models With Dictionary Learning*. Tech. rep. Anthropic. URL: <https://transformer-circuits.pub/2023/monosemantic-features>.

References ii

-  Bussmann, Bart, Patrick Leask, and Neel Nanda [Dec. 2024a]. *BatchTopK Sparse Autoencoders*. URL: <http://arxiv.org/abs/2412.06410> [visited on 04/02/2026].
-  — [Dec. 2024b]. “Learning Multi-Level Features with Matryoshka SAEs — LessWrong”. In: URL: <https://www.lesswrong.com/posts/rKM9b6B2LqwSB5ToN/learning-multi-level-features-with-matryoshka-saes> [visited on 05/10/2026].
-  Chanin, David, James Wilken-Smith, Tomáš Dulka, Hardik Bhatnagar, Satvik Golechha, and Joseph Bloom [Sept. 2024]. *A Is for Absorption: Studying Feature Splitting and Absorption in Sparse Autoencoders*. URL: <https://arxiv.org/abs/2409.14507>.
-  Cybenko, George [1989]. “Approximation by Superpositions of a Sigmoidal Function”. In: *Mathematics of Control, Signals and Systems* 2.4.
-  Elhage, Nelson, Tristan Hume, Catherine Olsson, Nicholas Schiefer, Tom Henighan, Shauna Kravec, Zac Hatfield-Dodds, Robert Lasenby, Dawn Drain, Carol Chen, Roger Grosse, Sam McCandlish, Jared Kaplan, Dario Amodei, Martin Wattenberg, and Christopher Olah [Sept. 2022]. *Toy Models of Superposition*. URL: <http://arxiv.org/abs/2209.10652> [visited on 07/17/2025].

References iii

-  Fillingham, Sean P., Andrew Gordon, Peter Lai, Xavier Poncini, David Quarel, and Stefan Heimersheim [Nov. 2025]. *SCALAR: Benchmarking SAE Interaction Sparsity in Toy LLMs*. URL: <http://arxiv.org/abs/2511.07572> [visited on 04/02/2026].
-  Johnson, William B. and Joram Lindenstrauss [1984]. “Extensions of Lipschitz Mappings into a Hilbert Space”. In.
-  Li, Kenneth, Oam Patel, Fernanda Viégas, Hanspeter Pfister, and Martin Wattenberg [June 2024]. *Inference-Time Intervention: Eliciting Truthful Answers from a Language Model*. URL: <http://arxiv.org/abs/2306.03341> [visited on 03/31/2026].
-  Lindsey, Jack, Adly Templeton, Jonathan Marcus, Thomas Conerly, Joshua Batson, and Christopher Olah [Oct. 2024]. *Sparse Crosscoders for Cross-Layer Features and Model Diffing*. URL: <https://transformer-circuits.pub/2024/crosscoders/index.html>.
-  Nakkiran, Preetum, Gal Kaplun, Yamini Bansal, Tristan Yang, Boaz Barak, and Ilya Sutskever [2019]. “Deep Double Descent: Where Bigger Models and More Data Can Hurt”. In: *arXiv preprint arXiv:1912.02292*.

References iv

-  Nanda, Neel [Mar. 2022]. “A Longlist of Theories of Impact for Interpretability”. In: URL: <https://www.lesswrong.com/posts/uK6sQCNMw8WKzJeCQ/a-longlist-of-theories-of-impact-for-interpretability> [visited on 03/31/2026].
-  Nanda, Neel, Andrew Lee, and Martin Wattenberg [2023]. “Emergent Linear Representations in World Models of Self-Supervised Sequence Models”. In: *Proceedings of the 6th BlackboxNLP Workshop: Analyzing and Interpreting Neural Networks for NLP*. Singapore: Association for Computational Linguistics. URL: <https://aclanthology.org/2023.blackboxnlp-1.2> [visited on 03/30/2026].
-  Olah, Chris, Nick Cammarata, Ludwig Schubert, Gabriel Goh, Michael Petrov, and Shan Carter [Mar. 2020]. “Zoom In: An Introduction to Circuits”. In: *Distill* 5.3. URL: <https://distill.pub/2020/circuits/zoom-in> [visited on 03/30/2026].
-  Panickssery, Nina, Nick Gabrieli, Julian Schulz, Meg Tong, Evan Hubinger, and Alexander Matt Turner [July 2024]. *Steering Llama 2 via Contrastive Activation Addition*. URL: <http://arxiv.org/abs/2312.06681> [visited on 08/12/2025].
-  Park, Kiho, Yo Joong Choe, and Victor Veitch [July 2024]. “The Linear Representation Hypothesis and the Geometry of Large Language Models”. In.

References v

-  Power, Alethea, Yuri Burda, Harri Edwards, Igor Babuschkin, and Vedant Misra [2022]. “Grokking: Generalization Beyond Overfitting on Small Algorithmic Datasets”. In: *arXiv preprint arXiv:2201.02177*.
-  Rajamanoharan, Senthoooran, Arthur Conmy, Lewis Smith, Tom Lieberum, Vikrant Varma, János Kramár, Rohin Shah, and Neel Nanda [Apr. 2024]. “Improving Dictionary Learning with Gated Sparse Autoencoders”. In.
-  SuperAnnotate [2023]. *Guide to Convolutional Neural Networks*. URL: <https://www.superannotate.com/blog/guide-to-convolutional-neural-networks>.
-  Templeton, Adly, Tom Conerly, Jonathan Marcus, Jack Clark, Yan Shlegeris, Trenton Bricken, Aidan O’Gara, Emmanuel Ameisen, Adam Jermy, Baobao Zhang, Chris Olah, and Sam McCandlish [May 2024]. *Scaling Monosemanticity: Extracting Interpretable Features from Claude 3 Sonnet*. URL: <https://transformer-circuits.pub/2024/scaling-monosemanticity/> [visited on 03/26/2026].